KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

PROJECT TITLE:- **Design and Implementation of Integrated SAP ERP System for Rekha Retail Pvt. Ltd.**

KIIT UNIVERSITY
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Bhubaneswar, Odisha, India

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1. Introduction

Enterprise Resource Planning (ERP) systems play a crucial role in integrating various business processes into a single unified system. SAP ERP is one of the most widely used ERP solutions that helps organizations streamline operations, improve efficiency, and ensure real-time data processing.

This project focuses on the implementation of SAP ERP for a fictitious company named **Rekha Retail Pvt. Ltd.**, which deals in food products such as sattu. The project includes the configuration and integration of three major SAP modules: Financial Accounting (FI), Material Management (MM), and Sales & Distribution (SD).

2. Problem Statement

In a growing retail business, managing financial transactions, procurement, and sales operations manually can lead to several challenges such as:

- Data inconsistency
- Lack of real-time visibility
- Delays in decision-making
- Poor inventory management

Therefore, there is a need for an integrated system that can automate business processes, ensure data accuracy, and improve operational efficiency.

3. Objectives of the Project

The main objectives of this project are:

- To design and implement an SAP ERP system for a retail company
- To configure the FI, MM, and SD modules
- To automate core business processes
- To demonstrate integration between different modules
- To generate real-time financial and operational reports

4. Company Overview

Rekha Retail Pvt. Ltd. is a fictitious company inspired by a real business idea. The company specializes in the production and distribution of food products, particularly sattu.

The company operates through:

- Procurement of raw materials
- Inventory management
- Sales through retail and online channels

The ERP system is implemented to manage all these operations efficiently.

3. Enterprise Structure

The enterprise structure defines the organizational hierarchy in SAP. It is essential for mapping business processes.

- **Client**: 100 (Top-level organizational unit)
- **Company**: Rekha Group
- **Company Code**: RR01 (Represents legal entity)
- **Plant**: RRPL (Manufacturing/Storage unit)
- **Storage Location**: RSL1 (Inventory storage)
- **Sales Organization**: RRSO (Handles sales activities)
- **Distribution Channel**: Retail / Online
- **Division**: Food Products

This structure ensures proper data organization and process flow.

4. Financial Accounting (FI)Module

The FI module is responsible for managing all financial transactions and reporting.

Configuration Steps:

- Creation of Company and Company Code
- Assignment of Company Code to Company
- Configuration of Chart of Accounts
- Setting Fiscal Year Variant

- Defining Posting Period Variant
- Creation of General Ledger (GL) Accounts
- Configuration of Tax Codes

Description:

The FI module records all financial transactions such as purchases, sales, and payments. It helps in generating financial statements like:

- Balance Sheet
- Profit and Loss Account

It ensures compliance with accounting standards and provides real-time financial data.

5. Material Management (MM) Module

The MM module handles procurement and inventory management.

Configuration Steps:

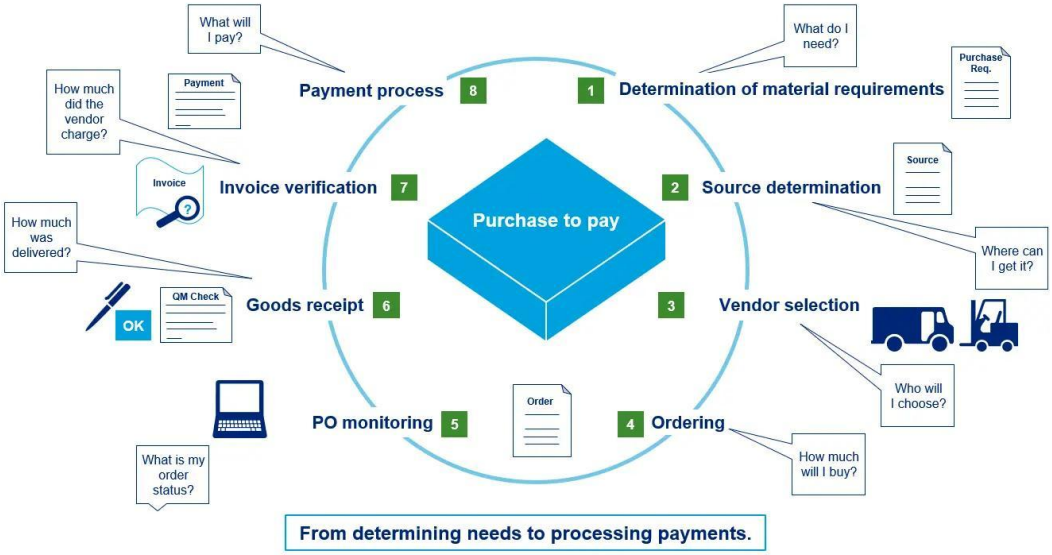
- Define Plant and Storage Location
- Assign Plant to Company Code
- Define Purchasing Organization
- Create Material Master Data
- Create Vendor Master Data
- Configure Pricing Procedure

Process Flow:

1. Purchase Requisition (PR)
2. Purchase Order (PO)
3. Goods Receipt (GR)
4. Invoice Verification

Description:

The MM module ensures efficient procurement of materials and proper inventory tracking. It helps avoid stock shortages and overstocking by maintaining accurate inventory records.



6. Sales and Distribution (SD) Module

The SD module manages sales processes and customer interactions.

Configuration Steps:

- Define Sales Organization
- Define Distribution Channel and Division
- Setup Sales Area
- Create Customer Master Data
- Configure Pricing Procedure

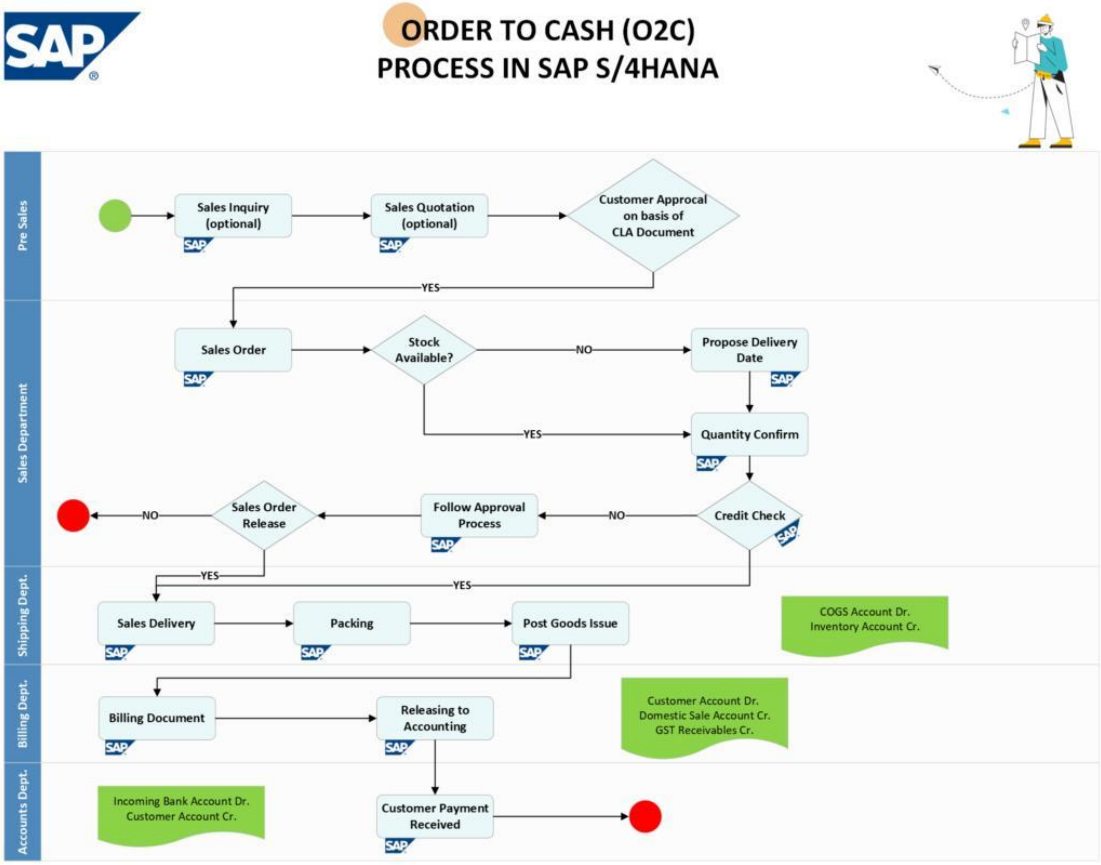
Define Sales Document Types

Process Flow:

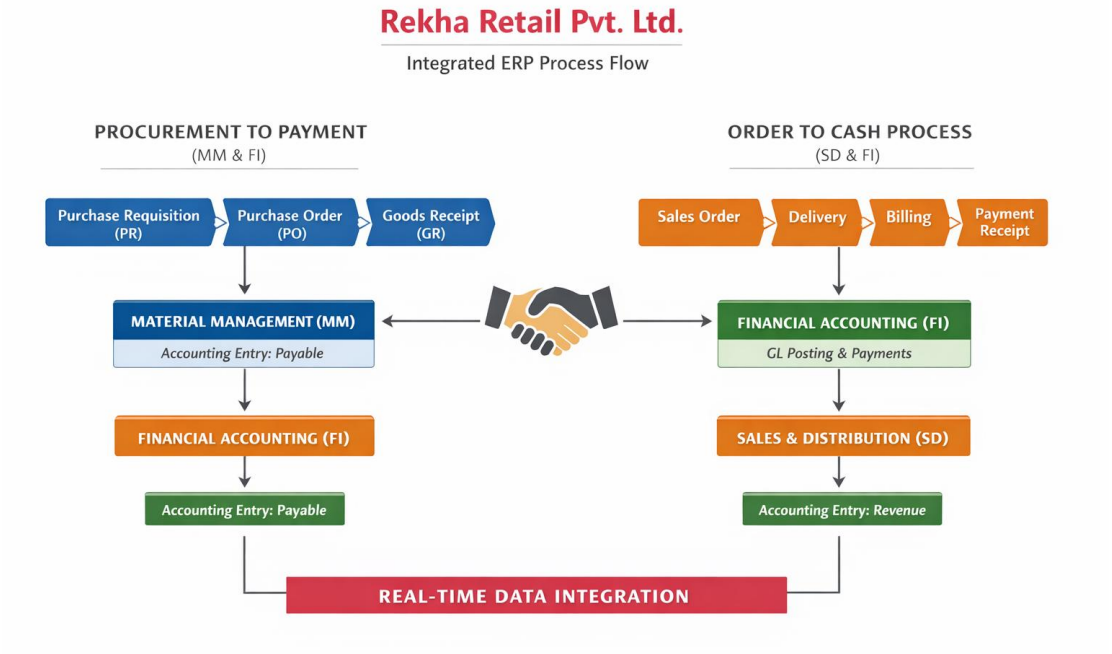
1. Inquiry
2. Quotation
3. Sales Order
4. Delivery
5. Billing

Description:

The SD module handles the complete sales cycle, from order creation to billing. It ensures timely delivery of goods and accurate billing of customers.



7. Integration of Modules



One of the key features of SAP ERP is integration between modules.

- **MM → FI**: Goods Receipt automatically updates financial accounts
- **SD → FI**: Billing generates revenue entries
- **MM ↔ SD**: Inventory data is shared between procurement and sales

This integration ensures consistency and eliminates data duplication.

8. System Workflow

```
                    REKHA RETAIL PVT. LTD.
                    ----------------------------
                    INTEGRATED ERP FLOW

(MM – PROCUREMENT)                    (SD – SALES)

[Purchase Requisition]                [Customer Inquiry]
        ↓                                     ↓
[Purchase Order]                      [Sales Order]
        ↓                                     ↓
[Goods Receipt]                       [Delivery]
        ↓                                     ↓
[Invoice Verification]                [Billing]
        ↓                                     ↓
        └──────→ [FINANCIAL ACCOUNTING (FI)] ←──────┘
                              ↓
                    [Payment Processing]
```

The overall system workflow is as follows:

- Procurement Process:
  Vendor → Purchase Order → Goods Receipt → Invoice → Payment
- Sales Process:
  Customer → Sales Order → Delivery → Billing → Payment

All these processes are interconnected and automatically update the financial records.

9. Screenshots Description

The project includes screenshots of:

- Company Code configuration
- GL Account creation
- Material Master and Vendor Master
- Purchase Order creation
- Sales Order processing
- Billing document

These screenshots provide visual proof of system implementation.

10. Technology Stack

- SAP ERP / SAP S/4HANA
- SAP GUI Interface
- GitHub for project hosting and version control

11. Unique Features

- Based on a real-world business idea (Rekha Sattu)
- Complete integration of FI, MM, and SD modules
- End-to-end automation of business processes
- Real-time data processing and reporting

12. Future Enhancements

- Implementation of SAP Fiori for improved user interface
- Integration with e-commerce platforms
- Use of AI for demand forecasting
- Advanced analytics using SAP tools

13. Conclusion

This project successfully demonstrates the implementation of an SAP ERP system for a retail business. The integration of FI, MM, and SD modules ensures seamless business operations, improved efficiency, and accurate data management. The project highlights the importance of ERP systems in modern organizations and provides practical exposure to SAP configuration and business process automation.